



musikal: Music recommendation system for DJ transitions

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The Goal

Problem: Typical music recommenders rank tracks by listener behavior, playlist co-occurrence and song metadata. None of this captures how a DJ might actually pick their next song. DJs listen for tempo, rhythm, energy, and harmonic structure to conclude whether two songs will actually blend well.

Goal: Given a library of N tracks and a query song s the DJ is actively playing, rank the remaining $N - 1$ tracks by how mixable each one is with the query, and return the top five by rank. We explicitly aim to use features derived from audio itself to intuit the next song to play.

Approach: We frame DJ mixability as a supervised learning problem trained on real DJ mixes based on a pair classifier trained on observed adjacent track transitions.. At recommendation time, a query track is scored against all candidate tracks, and the system returns the top five most mixable tracks from a defined music library (input corpus).

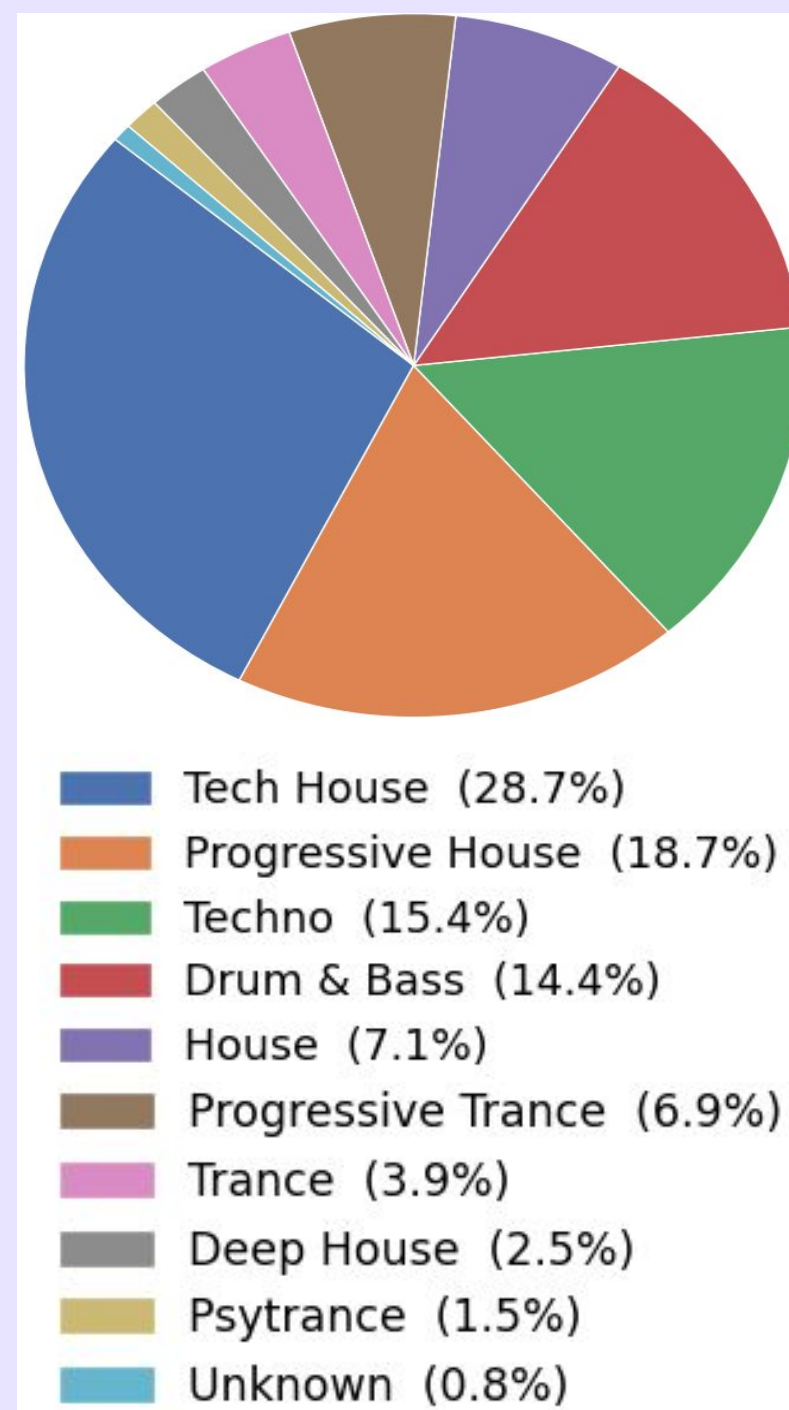
Dataset: DJ Mixes



5,040 DJ sets 64,754 transitions 63,038 tracks

Our subset: **3,827 tracks** (filtered from above to songs appearing in 3+ mixes, data manually scraped from Youtube)..

Genre Breakdown



Positive pairs: Two adjacents tracks from the table within the same mix

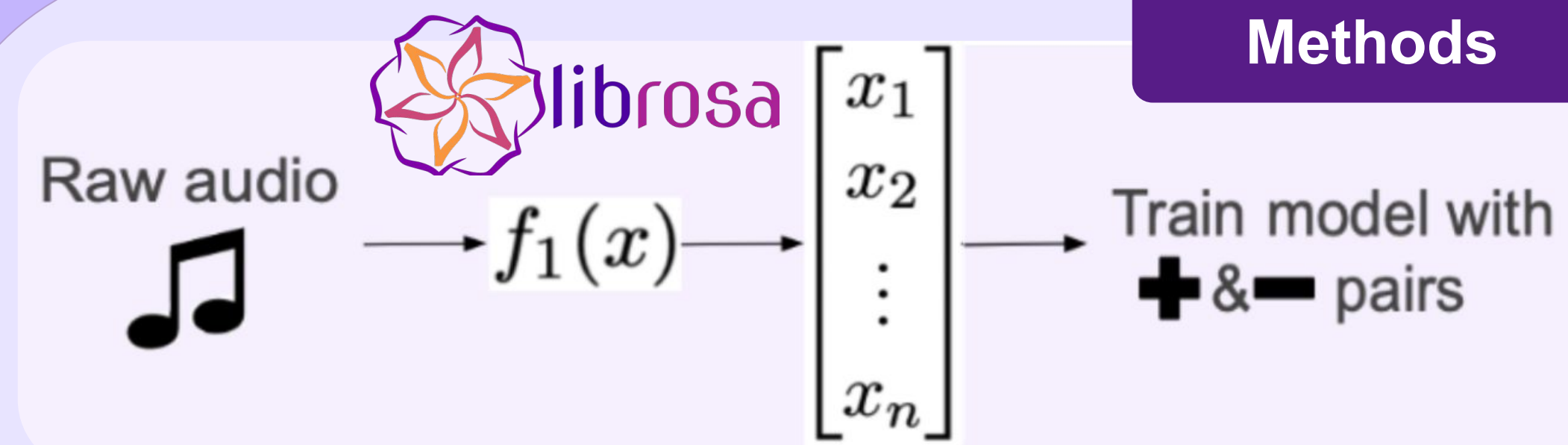
Negative Pairs: each positive pair randomly paried with non-adjacent track to act as a negative.

Mix #0421

[A] → [B] →
[C] → [D] →
[E]

positives: (A,B)
(B,C) (C,D) (D,E)
negative: (A, Z) Z
sampled randomly
from outside this
mix

Methods



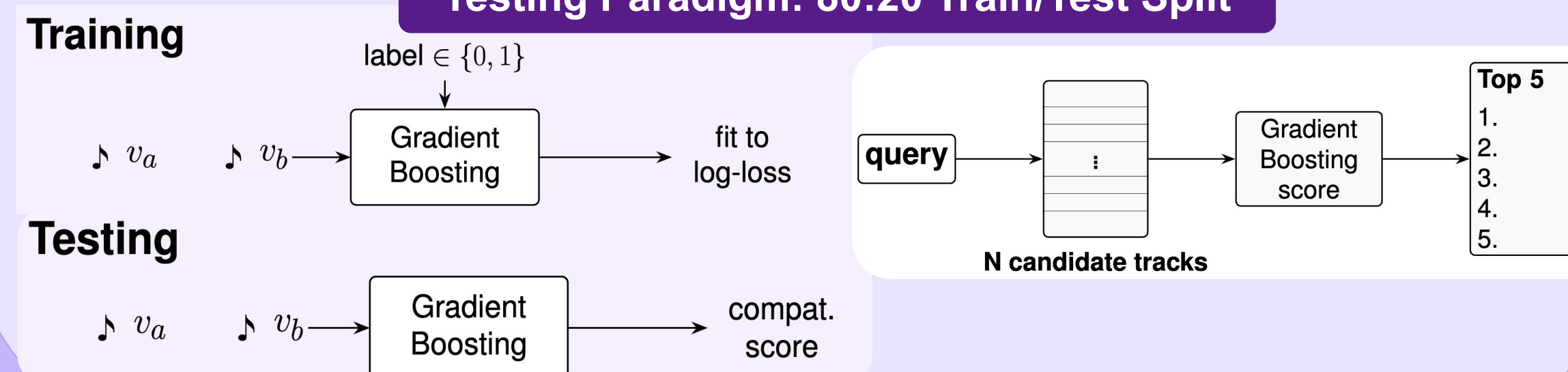
$f_1(x)$ Extract all features from librosa: **tempo, chroma, tempogram, MFCC...** → Integrate crafted features such as Harmonic BPM (below).

$$HBPM(T_A, T_B) = \min(|T_A - T_B|, |T_A - 2T_B|, |T_A - T_B/2|)$$

DJs mix at three rhythmically compatible speeds and our harmonic BPM variable captures these metrics..

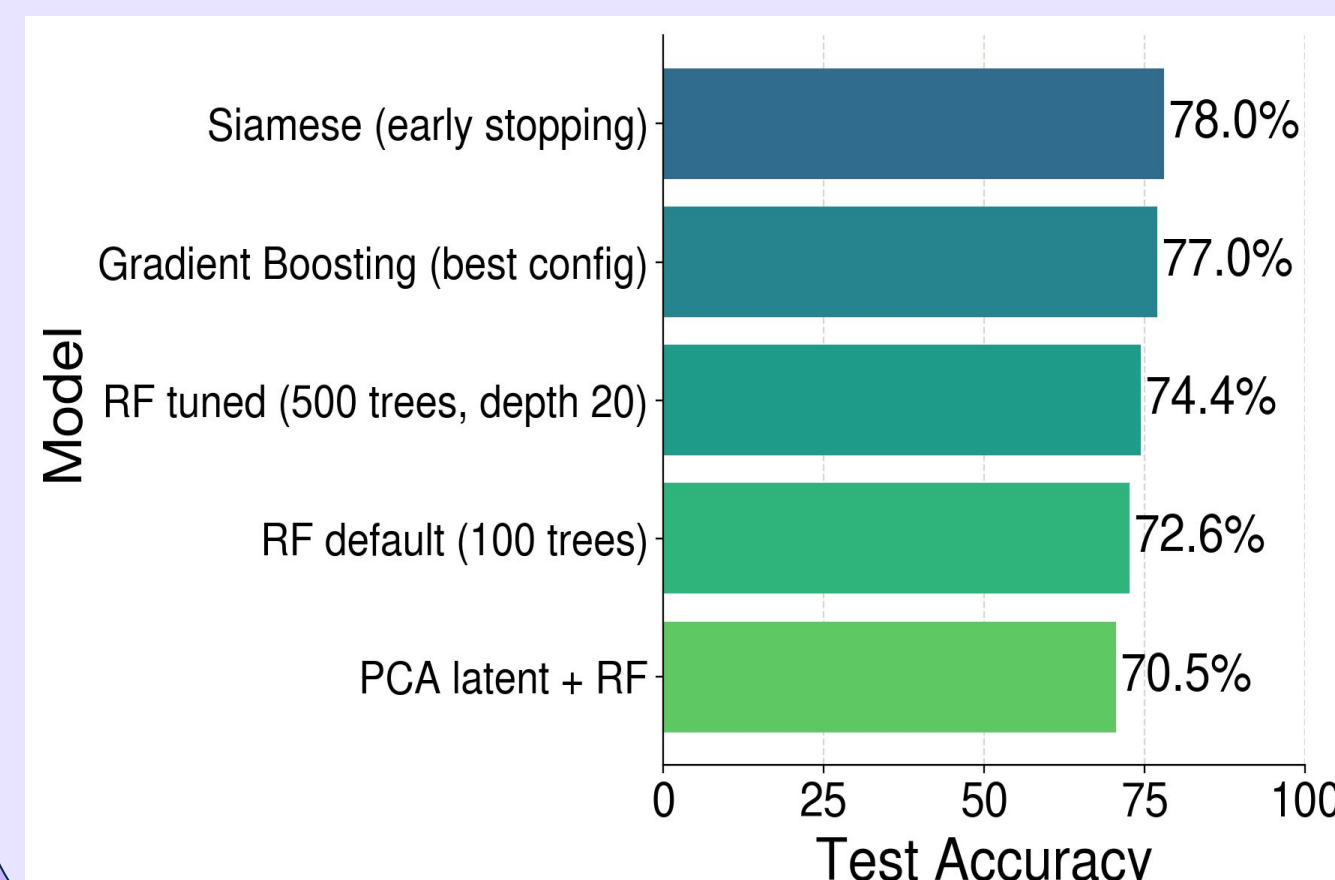
	Match	Double-time	Half-time
T_A (BPM)	128	140	90
T_B (BPM)	128	70	180
Plain $ T_A - T_B $	0	70	90
Harmonic HBPM	0	0	0

Testing Paradigm: 80:20 Train/Test Split



Model Choice

Model Accuracy Comparison



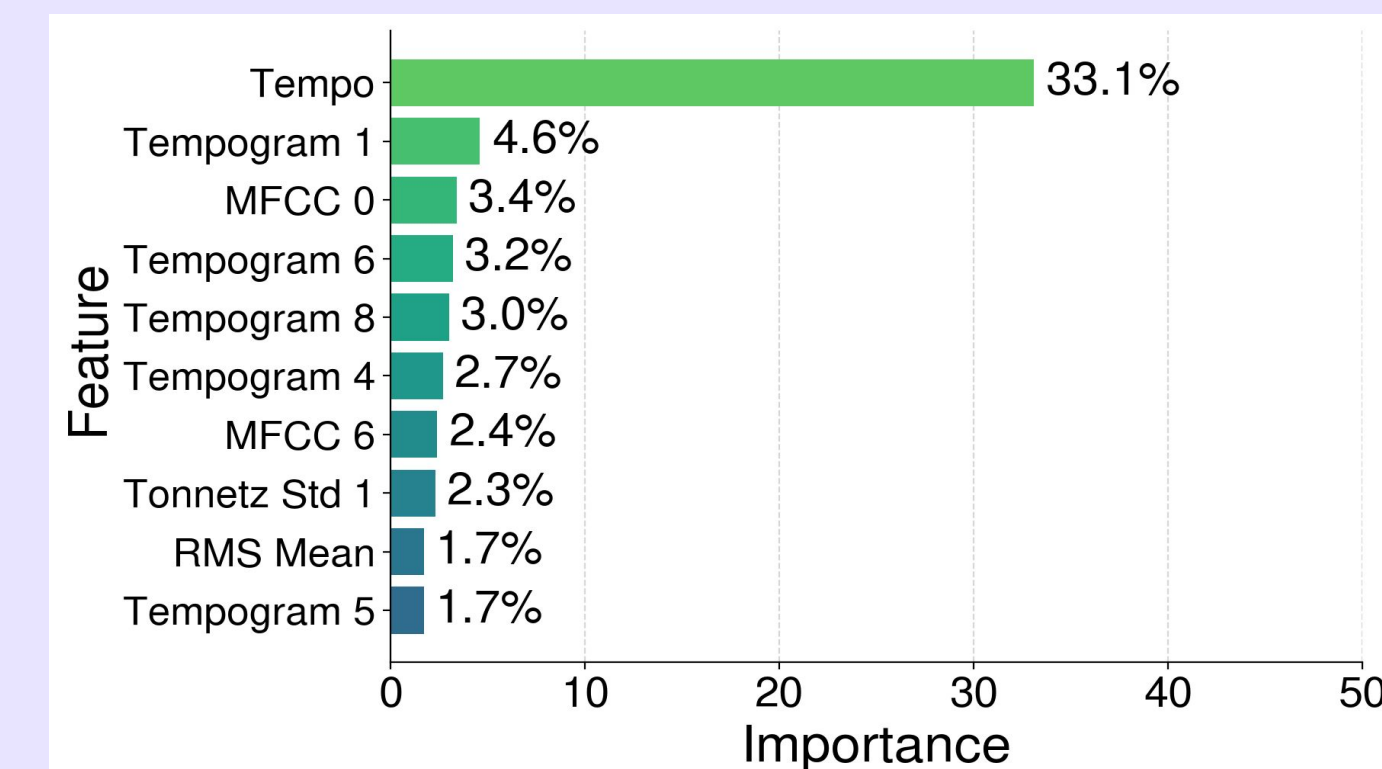
- **Random Forest** as a baseline which handles mixed feature and gives importances, tuned it with more trees and capped depth for stability,
- **Gradient Boosting** (200 trees, lr 0.1, depth 3) which won out by sequentially correcting errors.
- **PCA to 10 latents plus RF** held performance steady, showing redundancy wasn't hurting us.
- **Siamese contrastive net (3-layer tower, 16-dim embeddings)** allow the model learn its own pair representations instead of engineering them.

Limitations

- Coverage: Features extracted for 3,827 tracks, more data could have improved our results.
- Genre bias: Tech House dominates. Performance on hip-hop is unvalidated.
- Directionality: Symmetric formulation ignores intro/outro structure. We need more dataset.
- Negative Sampling: We will use harder, more realistic negative pairs in the future.
- Potential data mislabeling: We rely on the labels already provided in the dataset, some tracks may be mislabeled.

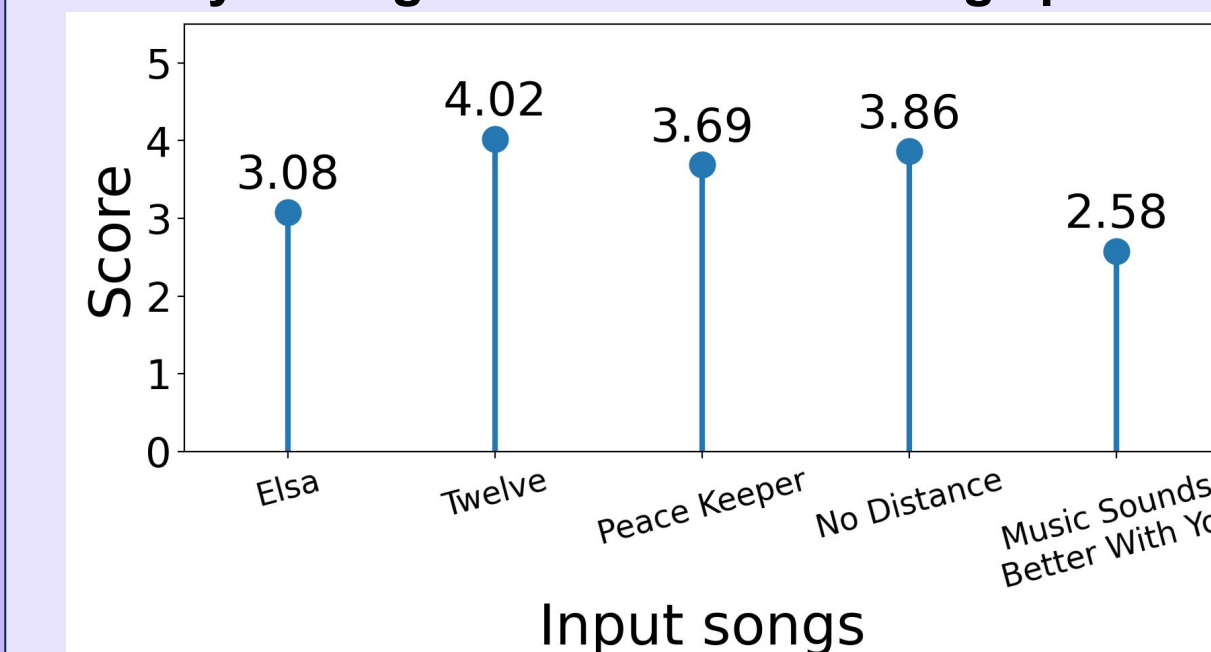
Results

Feature Importance

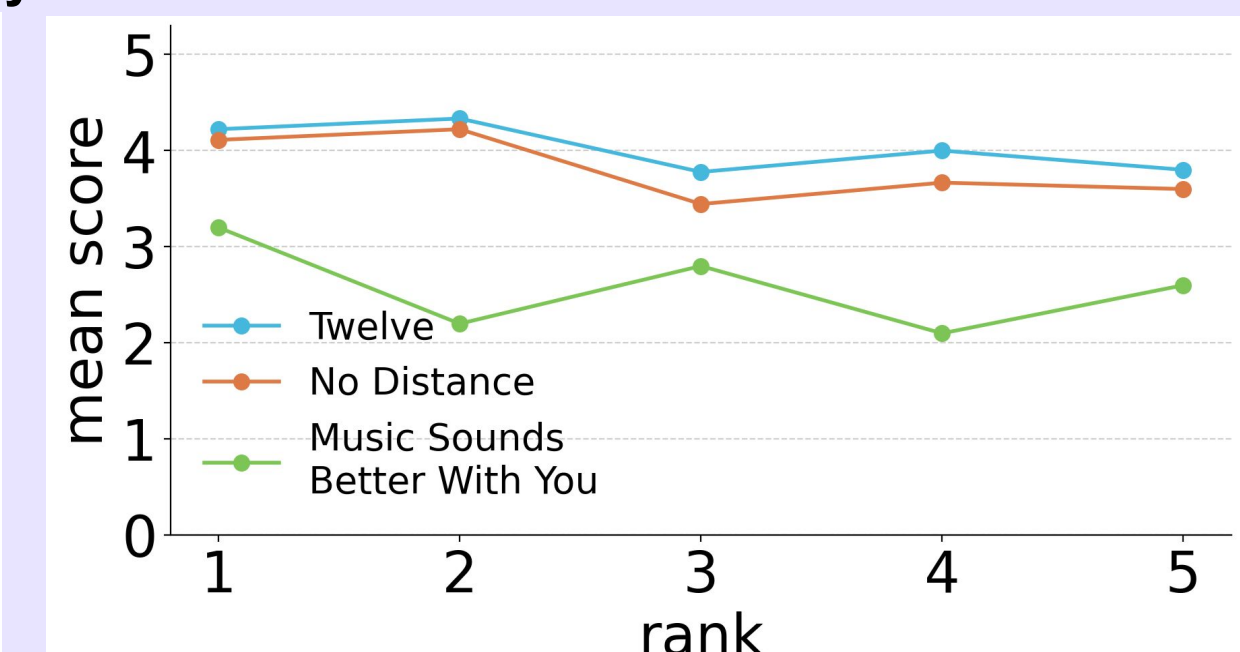


BPM lights up as most important feature predictably. Rhythmic structure indicates learning of beat patterns and matching. Timbral features indicate tracks are mixing with similar sound and textual patterns [MFCC].

Survey average of Five Ranked Songs per Query

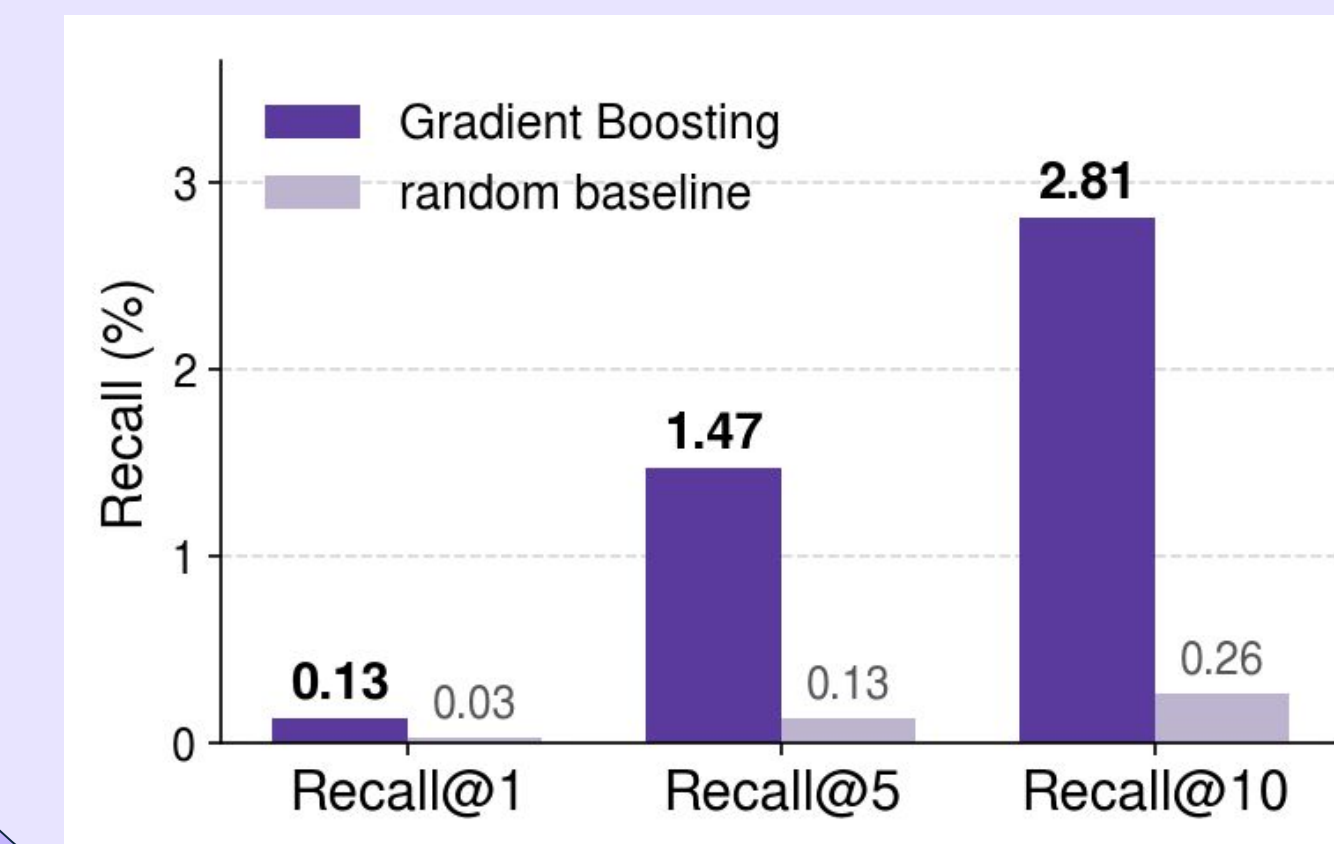


Average ratings after user survey landed between 2.58 and 4.02, with 70% satisfaction suggesting the model's recommendations generally felt musically coherent to listeners.



Average rating for each output song in the ranked order of 3 testing songs. Showing the rank is somehow coherent with its definition as the score decreases with the rank.

Next track retrieval: model vs. random



Recall@k against 3,827 candidate tracks. Gradient Boosting beats random selection by ~5x at k=1 and ~11x at k=10. Absolute numbers stay low because there is exactly one correct next track per query.

References

- [1] Aljanaki, A., Yang, Y., & Soleymani, M. (2017). Developing a benchmark for emotional analysis of music. PLOS ONE, 12(3), e0173392. <https://doi.org/10.1371/journal.pone.0173392>
- [2] ArXiv:2110.06525v1 [cs.LG]. (2021, October 13). AUTOMATIC DJ TRANSITIONS WITH DIFFERENTIABLE AUDIO EFFECTS AND GENERATIVE ADVERSARIAL NETWORKS.. Retrieved from <https://arxiv.org/pdf/2110.06525>
- [3] Teticio. (n.d.). DeeJ-AI: Create automatic playlists by using deep learning to listen to the music [Source code]. GitHub. Retrieved from <https://github.com/teticio/DeeJ-AI>